Practical Session 1 – Introduction to Data Analysis

Big Data Course – Third-Year Undergraduate (L3)



Learning Objectives

This first practical session aims to help you:

- become familiar with the working environment (`Python`, `Jupyter Notebook`);
- manipulate a real-world dataset;
- identify common data quality issues;
- perform initial data cleaning and simple exploratory analysis.

This practical session is **not graded**, but your work **must be submitted** on Community. It is an essential step in preparing for the rest of the course and the projects. The submission makes it possible to check your engagement in the learning process and provide individual feedback.

During practical sessions, we will use *Jupyter Notebook* with a *Python* kernel. For the first two practical sessions (TP1 and TP2), you may use the online environment:

`https://jupyter.org/try-jupyter/lab/`

⚠ Warning ⚠

This environment requires no installation and is easy to get started with. However, sessions are temporary, so files must be saved locally before closing the browser. You are strongly advised to save your work regularly and keep a local copy of your notebooks.

A recommended alternative is to install a local Python environment using *Miniconda*. This can be installed directly in your home directory and provides a persistent Jupyter environment. This solution is strongly recommended for subsequent practical sessions and projects.

Dataset

The dataset used for both practical sessions is available on Community. It is a real-world dataset designed to reflect situations commonly encountered in data analysis.

It consists of a table containing **499 tweets** about New Year's resolutions for 2015. Each record corresponds to a unique tweet and includes, among other things, the publication date and time, location information, the original tweet text, and the associated resolution category.

Like any real-world dataset, the provided data may contain:

- missing values;
- inconsistencies in format or content;
- outliers.

These imperfections are **intentional** and are an integral part of the assignment. The goal is to expose students to imperfect data and encourage them to apply appropriate cleaning and analysis methods.

Working Environment

The work must be completed in a **Jupyter notebook** using Python and the following libraries:

- `pandas`
- `numpy`
- `matplotlib` and/or `seaborn`

Tasks

1. Loading and Inspecting the Data

- Load the dataset into a Jupyter notebook.
- Display the first few rows of the dataset.
- Report the number of rows and columns.
- Identify the data types of the different variables.

2. Describing the Dataset

Briefly describe the dataset:

- the meaning of the main variables;
- the types of data (numerical, categorical, temporal, etc.);
- the presence of any missing values.

3. Data Cleaning

Perform an initial cleaning of the dataset:

- handle missing values (justified removal or replacement);
- correct or remove outliers where necessary;

- convert variable types if needed (dates, numbers, categories).

Briefly justify your choices in the notebook.

4. Simple Exploratory Data Analysis

Carry out an initial exploratory analysis:

- descriptive statistics (mean, median, minimum, maximum, etc.);
- simple visualizations (histograms, line plots, bar charts).

Graphs must be readable and accompanied by a short comment.

5. Saving the Cleaned Data

Export the cleaned dataset to a new file (for example, `CSV`), which will be reused in the next practical session.

Expected Deliverables

At the end of this practical session, you must submit:

- a clearly commented Jupyter notebook;
- the cleaned dataset file;
- a short conclusion (a few lines) summarizing the main observations.

Important Notes

- This practical session is not graded, but it directly prepares you for **TP2** and the projects.
- Clear code and comments are essential.
- Results must be interpreted, not merely displayed.